Statement of Purpose

Research Interests: Visualization, Data Science, Social Computing, Computer Graphics.

Data should be of the people, by the people and for the people. I am interested and experienced in using visualization to make socially produced data meaningful for understanding and improving society. And my ultimate goal is to enable everyone to analyze large-scale data freely with the power of visualization.

The strongest influence on my future goals has been my research for more than two years on data visualization in the Visual Analytics Group, State Key Lab of CAD&CG, under the supervision of Professor Wei Chen.

My first research project was about a Eulerian approach to analyze the crowd flow data extracted from mobile phones. The greatest challenge we faced was dealing with the low and inconstant accuracy of cell tower based location records. Our solution involved the use of the flow field in the Eulerian view, which focuses on specific locations, instead of the Lagrangian description. We designed a suite of visualization techniques to illustrate the dynamic evolutions of the flow over the networks. I contributed to the design and implementation of the visual analytics system, *Mobility Viewer*, which supports situation-aware understanding and visual reasoning about human mobility. This research led to my being included as the second author of a research paper, which was accepted by *IEEE Transactions on Intelligent Transportation Systems*.

Besides exploring crowd flow data, I am also interested in visualizing human relationships. A research project that I have been managing is *TieVis*. The idea behind this has been to establish an edge-based analysis framework for visualizing interpersonal ties so as to help users identify ties with similar trends, compare ties with different trends, and find hidden patterns. I personally designed and implemented most of the system, and we have used it to study dynamic networks in a dataset of chats between game players. I was the first author of a paper we submitted to *Workshop on Intelligent Data Analytics and Visualization 2016*. In addition, I gained valuable knowledge in another project in which for one month I surveyed large graph layout and rendering algorithms. That project was able to visualize a network of more than two thousand people in only one second.

A favorite topic in my research has been how to use visualization to help people explore the patterns and knowledge closely connected to their daily lives. From the cell tower based location record dataset in *Mobility Viewer* and the game player chat record dataset in *TieVis*, I have seen how data is ubiquitous in our modern lives and how human beings can benefit from good data analysis. Through my research, I have discovered a huge demand for data analysis experts, but I have also seen a great shortage of qualified people. Thus, it is my hope that one day I can help to develop a user-friendly visual analysis information system, so people can analyze large datasets as easily as they can edit documents on computers today.

I believe that pursuing a Ph.D. at the Georgia Institute of Technology is a great path for me to continue my research and pursue my goals. Besides a broad grounding in computer science due to its well-rounded curriculum, it attracts me with its powerful research team in visualization. I dream of working with Prof. Alex Endert and Prof. Polo Chau, who are both innovative and productive in combining visualization with data analysis technology, which is parallel to my research goals.

I also believe that I am well qualified to become a Ph.D. student because of my outstanding performance in academics, my professional experience and my tenacity in tackling heavy workloads.

Majoring in Computer Science and Technology at Zhejiang University, I have gained a solid foundation and good understanding of computer science. I have been a top student in the department, and I have taken Chu Kochen Honors College honors math courses like Linear Algebra, Probability, Mathematical Statistics, and Stochastic Processes. They have given me confidence to handle both mathematics-related courses and research. Furthermore, I am accustomed to reading and writing in English for all courses in the Computer Science Department.

Since the data scale is increasing in this Age of Big Data, I believe that my professional experience in Data Science and Computer Graphics work could also contribute to my research in visualization.

More than one year of experience administrating the 20-node computer cluster of the Visual Analytics Group exposed me to some practical concerns about managing big data. I was responsible for making the cluster run smoothly and dealing with assorted issues, such as designing the network topology, reconfiguring the operating system and deploying a Hadoop Distributed File System (HDFS). I also helped the group preprocess large datasets on the cluster with Hadoop and Spark. In the project, *Mobility Viewer*, I used Spark to independently process 14 billion trajectory records (about 1.8 terabytes) and 4.5 million detailed call records on the cluster. It was onerous to clean, transform and reduce them to meet all kinds of analysis needs, but the satisfaction of dealing with something closely related to real life was indescribable.

My knowledge and experience in Computer Graphics have been gained through courses, projects and competitions. I got a full picture of the rendering pipeline in the course, Computer Graphics, and then I delved into more topics, like mesh deformation, texture synthesis and precomputed radiance transfer, in the course, Advances in Computer Graphics. For the former course, I used OpenGL to implement a 3D game. In the latter course, I used DirectX and HLSL to implement an algorithm from a paper from *SIGGRAPH 2005* called “Precomputed Shadow Fields for Dynamic Scenes”. Thanks to these two projects, I gained enough skill to team up with several optical engineering students in *the 2015 Challenge Cup Competition*. We used OpenGL and 60 projectors to develop an autostereoscopic (glasses-free 3D) system.

The highlight of my professional experience has been working as a visiting intern in the HKUST Multimedia Technology Research Center to help develop a next-generation streaming cloud for high-quality multimedia broadcasting. This experience as well as my former research experience has honed my ability to grasp new knowledge in a short amount of time, communicate clearly with colleagues, and effectively implement a state-of-the-art prototype.

Finally, I believe I am qualified to become a Ph.D. student because of my capability and passion in handling and successfully completing heavy workloads. My experience working as a research assistant while maintaining a top class ranking has helped me develop strong skills in time management and in balancing my workload so as to handle stress and fatigue and maximize efficiency. It has also given me confidence to face even greater challenges and maximize my potential to reach my life goals.

Thank you for considering my application.